

NOISEMAP · REQUIREMENTS

NoiseMap requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/noisemap		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Status figures from NSM-CAL-001 v0.3 (constructable design, NSM-DDR-003); R13 stated against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3; R13 redefined to the NoiseMap parts under NSM-DDR-001 D1; R2 metric set adopted (D5); status column from NSM-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

NoiseMap requirements

These requirements are targets for concept review and must be revised with a co-design partner before the design is frozen. The metric set (R2) and the budget scope (R13) follow NSM-DDR-001 D5 and D1; the relaxed mass limit (R10), the wind flag method (R11) and the interval rule (R12) follow NSM-DDR-002. All were decided by Amish, 2026-09-25: go with recommendation. Status is taken from the TRL 3 calculation note NSM-CAL-001 v0.2; "met on paper" means met by calculation only. None is not met; three are at risk (R3, R4, R5), four can be settled only by test (R8, R9, R11, R14), and eight are met on paper or by design.

Design case: one node on a 114.3 mm street pole 0.45 m behind the curb, microphone 4.0 m above the sidewalk and 0.45 m from the pole face, in a mixed street with traffic by day and bars at night.

Table 1. Requirements.

ID	Requirement	Target	Verification	Status (NSM-CAL-001)
R1	Privacy: no audio samples are stored or leave the microphone head; only levels cross the cable or the radio	No data storage in the head; cable carries levels only; head firmware published with a build hash and read-out protection	Design review, firmware audit, cable capture	Met by design; the guarantee rests on the firmware, since the cable could carry a speech codec (NSM-DDR-001 D2)
R2	Reported metrics	Per 15 min: fifteen 1-minute LAeq, LAeq, LAFmax, LAFmin, L10, L90, LCEq, device status	Design review	Met by design (22 bytes per record)
R3	Accuracy of LAeq,15min against a co-located class 1 meter	Within ± 2 dB from 40 to 100 dBA, after field calibration	Side-by-side field test	At risk: expanded uncertainty 2.9 dB on assumed terms; facade bias up to 2.4 dB before a site correction
R4	Measuring range	35 to 110 dBA	Bench test with calibrator and reference source	At risk: ICS-43434 34.9 to 113 dBA (no margin); IM72D128 27.9 to 113 dBA
R5	Frequency weighting	A and C weighting as defined in IEC 61672-1, within class 2 tolerances from 63 Hz to 8 kHz (goal)	Bench test against a reference meter	At risk: roll-off equalized to ± 0.8 dB at 63 Hz; membrane, windscreen and head diffraction unknown
R6	Time stamps	Each record time-stamped within 2 s of UTC	Design review (LoRaWAN network time)	Met by design
R7	Energy	Energy neutral at 1.5 peak sun hours; 14 days without sun	Energy budget, bench test	Met on paper (5.81 Wh/day for 0.29 Wh/day; 53 days)
R8	Ingress and weather	IP65 enclosure; microphone head survives driving rain; -20 °C to +50 °C	Spray test, thermal chamber	Not verifiable at TRL 3
R9	Calibration check	Field check with a 94 dB, 1 kHz calibrator in 10 min or less; offset stored on the node	Timed trial	Not verifiable at TRL 3
R10	Installation and mass	Two trained people, 45 min, no drilling or pole wiring; 60 to 140 mm poles; 3.5 kg or less (relaxed from 3 kg, NSM-DDR-002); withstands 35 m/s gusts	Timed trial, weighing, bracket calculation	Met on paper on mass (3.45 kg for the constructable design; margin 0.05 kg); wind met on paper; fit met by design; time not verifiable at TRL 3
R11	Contaminated data	Intervals with wind above about 5 m/s or heavy rain are flagged, wind by a level-based flag (no added sensor) and rain on the server from weather data (NSM-DDR-002)	Field comparison with a weather station	Not verifiable at TRL 3: method chosen; the threshold needs field data (TRL 4, on hold)
R12	Radio use	Within EU868 1 % duty cycle and The Things Network fair use (30 s uplink per day); 15-minute interval	Airtime calculation	Met on paper: 23.7 s/day at SF9; 29.6, 29.6 and 30.0 s/day

ID	Requirement	Target	Verification	Status (NSM-CAL-001)
		at SF7 to SF9, lengthened automatically at SF10 to SF12 (NSM-DDR-002)		at 24, 48 and 87 min for SF10 to SF12
R13	Parts cost	NoiseMap parts (lines 7 to 14 of the BOM) at or under the \$100 value-engineering target per node; the FieldNode core is costed in the FieldNode repo and the full node cost is stated	Priced BOM	Met on paper: NoiseMap parts \$72.00, \$28.00 under the target; full node \$211.00 with the \$139.00 FieldNode core
R14	Service life	5 years outdoors, with a yearly windscreen change and one battery change	Design review, UV-stable materials	Not verifiable at TRL 3
R15	Openness	Hardware, firmware, calibration method and data format published under the repo licenses	Design review	Met by design

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Microphone height	about 4 m above the ground	4.0 ± 0.2 m assessment height for EU strategic noise maps (Directive 2002/49/EC, Annex I)
Quietest night level to resolve	about 35 dBA	Quiet residential street at night (estimate)
Loudest level to resolve	about 110 dBA	Close pass of a loud motorcycle or siren (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Wind speed that spoils readings with a foam windscreen	about 5 m/s	Common practice (estimate, to be checked)
Microphone data	ICS-43434: 65 dB SNR, 120 dB SPL overload, 60 Hz to 20 kHz; IM72D128: 72 dB(A) SNR	ICS-43434 product page (TDK InvenSense); IM72D128 from the DNMS README (DNMS)
Street pole diameter	60 to 140 mm, design case 114.3 mm	Common lighting and signal poles (estimate)

Safety: Requirements R8 and R10 involve work at height beside traffic and a lithium cell. See the safety section of NSM-PRC-001.